

Mohammad Bilal

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EDUCATION

FAST NUCES, Chiniot Faisalabad (CFD)

Bachelor of Science in Computer Science

Expected Graduation: June 2026

SKILLS

Languages & Frameworks

Proficient: JavaScript, HTML, CSS, C, C++ (3 years), Python (2 years)

Intermediate: React.js, Node.js, SQL

Beginner: Next.js, TypeScript, C#

Software & Platforms

LangChain, GROQ Inference, Ollama, Jupyter Notebook, Bootstrap, Microsoft Office (Excel, Word, PowerPoint)

AI & DevOps

Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Neural Networks

Tools & Frameworks

Clerk Auth, Prisma, Tailwind CSS, tRPC, REST APIs, MongoDB, PostgreSQL, Hugging Face, FastAPI, Python Flask, TensorFlow, PyTorch, DeepFace, Streamlit, Tesseract OCR, PaddleOCR

EXPERIENCE

Backend Developer Intern – Smart Home Energy Management System

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Oct 2024 – Dec 2024

- Developed backend APIs using Node.js and Express to control IoT devices for home automation.
- Integrated MongoDB for logging, tracking, and managing device activity data.
- Focused on energy analytics, cost efficiency, and secure user access using JWT.

AI Research Intern – Independent Projects & Open-Source

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Jul 2024 – Present

- Built and published AI apps using LangChain, Gemini APIs, and vector databases like Pinecone.
- Implemented RAG (Retrieval-Augmented Generation) pipelines for code understanding and financial insights.
- Optimized prompts, tuned LLMs, and integrated analytics dashboards for AI-driven platforms.

PROJECTS

Facial Recognition Web App | AI + Deep Learning (~30 hours)

Feb 2025 – Mar 2025

- Built a face recognition web app using Flask and DeepFace with Gemini API integration.
- Enabled image upload, face comparison, and matching confidence visualization.
- Supported ZIP uploads, image previews, and secure downloads via Flask and Werkzeug.

AI-Powered Text Summarizer | Full Stack AI App (~50 hours)

Jun 2024 – Jul 2024

- Developed a summarization app with FastAPI backend and Hugging Face transformers.
- Frontend built using Next.js and TypeScript for a sleek, responsive UI.
- Supported long-form content summarization with real-time output from LLMs.

Conversational RAG Search Engine | LangChain, Gemini, Llama3, Pinecone (~80 hours)

Dec 2024 – Mar 2025

- Built a multi-source, agent-based AI search engine with LangChain, Streamlit, and RAG pipelines for querying web, Wikipedia, ArXiv, and custom documents.
- Integrated Gemini API and Llama3 (**via Groq**) to generate conversational responses using chat history and semantic search.
- Used Pinecone for vector storage and retrieval, enabling contextual memory and personalized query handling.

Management Systems Suite | C++, C# forms, TypeScript, SQL (~90 hours)

Jun 2023 – Sept 2023

- Developed academic, hospital, library, and airline management systems using C++, C#, TypeScript (Next.js), and Oracle SQL.
- Implemented features like user roles, scheduling, inventory tracking, and database integration for CRUD operations.
- Focused on modular design, data validation, and user-friendly interfaces across all systems.

AI & Logic-Based Game Collection | Python, C++, AI (~60 hours)

May 2022 – Jun 2025

- Built multiple games including **Tic Tac Toe, Bingo, Checkers (AI), and Sudoku Solver** using OOP and AI algorithms like minimax, backtracking and CSP (Constraint Satisfaction Problems).
- Applied constraint satisfaction and heuristics in Python-based games to simulate intelligent play.
- Designed clean CLI/GUI interfaces and emphasized game logic accuracy and replayability.

Activities

Member of FAST Society of Cyber Security - Lead the Technical Society in Domain of Cyber Security
Photographer FPS NUCES - Collaborate with Media Team for Coverage in Events Workshops & Conferences

Oct 2022 – Present

Nov 2022 – Present